



Fig. 2: Solderless breadboard

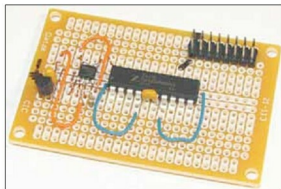


Fig. 3: Generic PCB

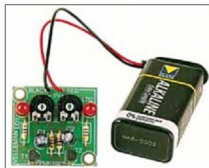


Fig. 4: Prototype on a custom-built PCB

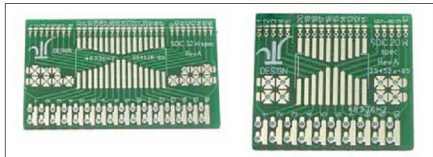


Fig. 5: SMD adaptors

able for high-voltage and/or high-current circuits. Although breadboard is an essential facet of the design process, the reliability and endurance of breadboard-based prototypes have been a big problem.

Printed circuit board wiring. Using printed circuit board (PCB) is a costlier but reliable method of prototyping electronic circuits. The most common PCB type widely used for hobby prototyping is the generic PCB, which has a pattern of holes and connections (multiple holes per pad) similar to a solderless breadboard.

Generic PCBs are easier to use than plain perforated boards, especially for IC-based circuits. But layout constraints are a minor drawback and prototyping may be much more difficult.

The 'real' PCB (which can handle virtually any component and power level)

used ubiquitously in the production of electronics is inappropriate for low-budget basic prototyping, because laying out the customised PCB and getting it fabricated may take time. Besides, it is very difficult to make changes or modifications in the initial circuit structure after finishing the prototype.

Using surface-mount devices (SMDs) instead of the traditional through-hole components makes a prototype more compact. The PCB layout becomes easier

because different things can be done on each side. At the same time, prototyping becomes much more difficult (requires high soldering skill), which calls for the use of special SMD prototyping boards/custom PCBs and handling/soldering tools.

An alternative way to set up the SMD prototype in 'dead bug' style is to use special DIP/SIP adaptors that have pads to solder SMD ICs. 'Surfboards' with solder pads for connecting larger wires are also available. These make it easy to wire a hybrid electronic circuit (through-hole plus SMD components) on a standard breadboard.

Prototyping difficulties and bugs

Electronics prototypes have many disadvantages and difficulties. Foremost among these is the time taken to build them. It can also be difficult to estimate the time required to complete a working prototype, particularly because they tend to be non-functional until they are almost complete.

Component selection and circuit design (pre-existing modules, breadboard circuits, custom-designed PCBs, etc) constrain the flexibility of the prototype, even as its functioning begins to disclose new possibilities. Another obstacle to adjusting the design of a prototype is that any change risks breaking it completely. This points to one of the other weaknesses of electronics prototypes: their inflexibility and fragility. Electronics prototypes are difficult to transport and maintain. This greatly limits their lifespan and scope.

Another problem with electronics prototypes is the bugs. It is an awkward task to find bugs on a prototype, as there are so many reasons for errors—from wrong wiring of breadboard components, loose connections, broken parts and problems with power supply, to errors in the design concept.

Examples of some bugs to expect are:

1. Design/schematic flaws: Poor design concept/drawing error in schematic diagram
2. Board layout problem: Wrong/